Week 8, Lecture 1 WOD: Answer Key

How Much Did the Prior Matter?

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. 2a is open-ended and gets a sample answer. All three panels are Beta-binomial (the setup from Week 7 Wednesday) with the same data, 21 of 30 tracked juveniles surviving, so the shared likelihood peaks at 0.70. Students don't see the data or any of these numbers.

Analyst	Prior	Prior worth (tortoises)	Prior mean	Posterior mean	95% credible interval
A	Beta(2, 2)	4	0.50	0.68	0.51 to 0.82
B	Beta(46, 4)	50	0.92	0.84	0.75 to 0.91
C	Beta(10, 10)	20	0.50	0.62	0.48 to 0.75

A Beta(α, β) prior is "worth" about $\alpha + \beta$ animals, against 30 tracked juveniles in the data.

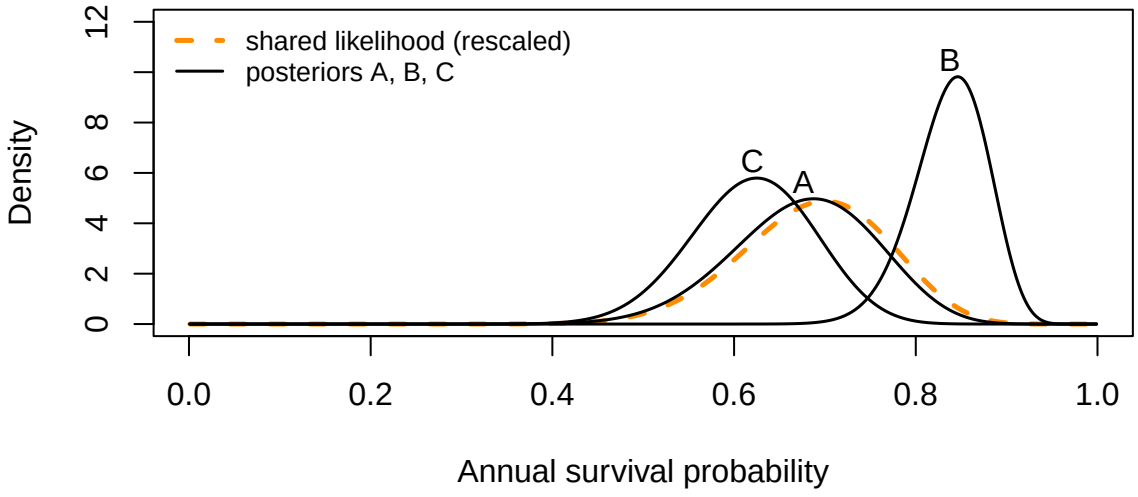
Use the debrief to lead into the germination example: the supplier prior there is worth 200 seeds against 40 seeds of data, a stronger prior relative to the data than Analyst B's.

Part 1: Find the likelihood

1a.

> Sketch the shared likelihood on the bottom panel (the height doesn't matter; get its peak and width roughly right). Which analyst's panel did you rely on most, and why?

Answer. A curve peaking near 0.70 and about as wide as Analyst A's posterior (roughly 0.5 to 0.85). Rely on Analyst A: with a nearly flat prior, A's posterior is almost the likelihood itself.



Why. The posterior is a compromise between prior and likelihood. When the prior is nearly flat it barely pulls, so the posterior is close to the rescaled likelihood. The other two panels check the sketch: B's posterior sits between its prior (near 0.92) and 0.70, and C's sits between its prior (0.50) and 0.70.

Watch for. Averaging the three posteriors (that bakes in B's and C's priors). Drawing the likelihood at B's posterior peak because B's curve is the tallest and narrowest. Students who use B or C should still notice the likelihood must sit on the far side of the posterior from that analyst's prior.

1b.

> Rank the three analysts' results from most to least dependent on their choice of prior. Give a one-line reason for each.

Answer. B (most), then C, then A (least).

- **B:** posterior pulled far from the likelihood toward 0.92; its prior is worth more than the data (50 vs. 30 tortoises).
- **C:** posterior pulled partway toward 0.50; a moderate prior (worth 20).
- **A:** posterior nearly matches the likelihood; a weak prior (worth 4).

Why. Every analyst used the same data, so the differences among posteriors come entirely from the priors. How far each posterior sits from the likelihood measures how much its prior mattered.

Watch for. Ranking by posterior width. B's posterior is the narrowest, but that narrowness comes from the prior, not from more data (all three had the same data).

Part 2: Which number?

2a.

> The three posterior means for annual survival are 0.68 (A), 0.84 (B), and 0.62 (C). A manager needs one survival rate for a population model. What would you tell them, and what would you report?

Sample answer. Don't use Analyst B's result. Its prior comes from juveniles in a predator-proof pen, and predation is a major cause of juvenile death in the wild, so the pen tells us little about this site. Report Analyst A's posterior (mean 0.68, 95% credible interval 0.51 to 0.82) as the main estimate, and note that a drought-year prior (C) pulls it down to 0.62. Say plainly that 30 juveniles aren't enough to make the choice of prior irrelevant.

Why. The same data give posterior means from 0.62 to 0.84, a spread that could decide whether a population model projects growth or decline (juvenile survival drives recruitment). That is a prior sensitivity analysis, and the right response is to report it and justify the prior, not hide it. Whether a prior is defensible is an ecological question about where it came from, not a statistical one.

Watch for. The handout deliberately doesn't ask whether the pen prior is realistic; let students raise it. If nobody does, ask "where did each analyst's prior come from, and does that population resemble this one?" before giving the answer. Other common answers:

- "Use B, it's the most certain." B's narrow interval comes from its prior, which is the least relevant one.
- "Average the three." Averaging doesn't fix an unjustified prior; it just lets the pen data in at a third of the weight.
- "Use C to be cautious." Reasonable only if this was a drought year at a site like C's; otherwise it's a different kind of irrelevant prior. A good answer can mention C as a sensitivity check.